

Handling Missing Data

Q1. What are the most common reasons for missing data in ETL pipelines?

Answer :-

- **Data entry errors** (users leave fields blank)
- System failures during data collection
- Integration issues between multiple data sources
- Schema mismatches or transformation errors
- Optional fields left blank
- Data corruption or loss during transfer

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Answer :-

- Leads to loss of important information
- Can introduce **bias** in the dataset
- Reduces dataset size → impacts model performance
- Missing data might contain **patterns** useful for analysis

Q3. Explain the difference between:

Listwise deletion

Column deletion

Also mention one scenario where each is appropriate.

Answer :-

Listwise Deletion :-

- Removes entire row if any value is missing
- Use when dataset is large and missing values are few

Column Deletion :-

- Removes entire column if many values are missing
- Use when a column has **too many missing values** and is not important

Example Scenarios:-

- Listwise: Removing customers with incomplete records in a large dataset
- Column: Dropping a column like “Middle Name” with 90% missing data

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Answer :-

Handling Missing Data

- Median is **robust to outliers**
- Mean gets affected by extreme values
- Example: Income data is usually skewed → median gives better central value

Q5. What is forward fill and in what type of dataset is it most useful?

Answer :-

Forward fill (FFill) replaces a missing value with the most recent previous non-missing value.

It is most useful in :-

- Time-series data
- Sensor reading
- Stock prices
- Daily weather or sales records

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Answer :-

Flagging creates an indicator (e.g., Income Missing = 1) before filling missing values.

Benefits :-

- Preserves information that a value was originally missing.
- Allows analysts/models to learn whether missingness itself is meaningful.
- Improves transparency and debugging in ETL.

Q7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

Answer :-

Missing income can reveal important patterns, such as:

- Customers may be **unwilling to disclose sensitive financial information**.
- Certain customer groups may consistently avoid providing income.
- The application or registration process may be too complicated.
- Missing income could indicate lower engagement or incomplete profiles.

Business insight: Instead of treating missing income only as a data quality issue, the company can use it to improve customer forms, identify customer behavior patterns, or create targeted marketing and data collection strategies.

Handling Missing Data